***NOTE***

Predictive digital twins can transform scheduled or reactive maintenance into proactive optimization, fundamentally changing how organizations manage their physical assets and unlock new value-added services. Companies can now offer predictive maintenance subscriptions, performance guarantees, or outcome-based contracts, turning operational insights into revenue streams.

### *REAL WORLD EXAMPLE: PREDICTING EQUIPMENT FAILURE*

Imagine an industrial pump equipped with a temperature sensor. Over time, it starts showing rising temperatures. A predictive digital twin monitoring this pump might be using a threshold-based rule (for example, "if the temperature exceeds 60 degrees Celsius, then trigger a warning") or a more advanced anomaly detection model trained on patterns from pumps that previously failed. The trained machine learning model knows that when the temperature of the pump has exceeded 60 degrees for more than two hours previously, this has caused a failure in the bearing. The goal is to forecast problems before they happen, avoiding costly breakdowns and unplanned downtime.

### *REAL WORLD EXAMPLE: PREDICTING TRAVEL TIME*

One of the most widely used predictive digital twins is likely in your pocket right now. Open Google Maps, and it will estimate how long it’ll take to drive anywhere you choose, not just based on current traffic, but on what traffic is likely to look like 30 minutes from now, as shown in figure 1.8. How does it do that?